SET

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Equilibrium uniqueness and representation

In nonlinear models, standard sufficient conditions for uniqueness can be difficult to verify.

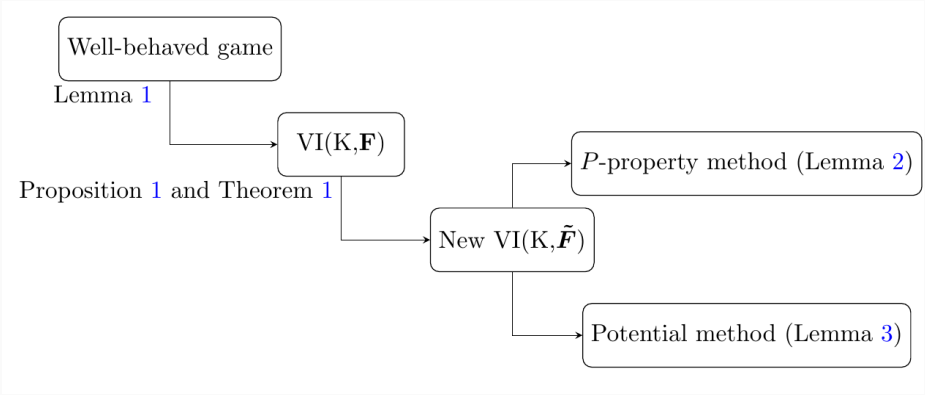
Some of this difficulty may come from how the equilibrium conditions are written.

The central question

How can we transform nonlinear equilibrium to simplify analysis, without changing the equilibrium set?

The paper develops sign-equivalent transformations (SET) to reveal useful mathematical structure in equilibrium problems.

The paper's map



An equilibrium is a best-response fixed point. The figure follows its VI representation through a sign-equivalent transformation to two methods for studying uniqueness.

Example 1: a two-player game

Two players choose $x_i \geq 0$, so $K = \mathbb{R}^2_+$.

Payoffs

$$u_1(x) = x_1\left(1 + \tfrac{1}{2}x_2\right) - \tfrac{1}{2}x_1^2,$$
$$u_2(x) = 4\left[x_2\left(1 + \tfrac{1}{2}x_1\right) - \tfrac{1}{2}x_2^2\right].$$

Each payoff is strictly concave in the player's own action.

Best responses

$$B_1(x_2) = 1 + \tfrac{1}{2}x_2, \qquad B_2(x_1) = 1 + \tfrac{1}{2}x_1.$$

The factor 4 changes player 2's payoff units, but leaves the best response unchanged.

Example 1: the VI and its transformation

For a candidate $x^* \in K$, use the negative marginal payoffs:

$$F_i = -\partial_i u_i, \qquad F(x) = \begin{pmatrix} x_1 - 1 - \frac{1}{2}x_2 \\ 4(x_2 - 1 - \frac{1}{2}x_1) \end{pmatrix}.$$

$$x^* \in \mathrm{NE} \iff \langle y - x^*, F(x^*) \rangle \geq 0 \quad \text{for every } y \in K.$$

SET: divide the second component by 4

$$\widetilde{F}(x) = \mathrm{diag}(1, \tfrac{1}{4})F(x) = \begin{pmatrix} x_1 - 1 - \frac{1}{2}x_2 \\ x_2 - 1 - \frac{1}{2}x_1 \end{pmatrix}.$$

Positive scaling preserves every component's sign on K. Since K is rectangular,

$$\mathrm{Sol}(K, F) = \mathrm{Sol}(K, \widetilde{F}).$$

Example 1: the two uniqueness routes

The transformed system has a symmetric derivative:

$$H = D\widetilde{F} = \begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{pmatrix}, \qquad \lambda(H) \in \{\tfrac{1}{2}, \tfrac{3}{2}\}.$$

P-property method

The principal minors of H are $1, 1, \frac{3}{4} > 0$.

The P-matrix condition gives at most one VI solution.

Potential method

$$\Phi(x) = \tfrac{1}{2}(x_1^2 + x_2^2) - \tfrac{1}{2}x_1x_2 - x_1 - x_2.$$

$\nabla\Phi = \widetilde{F}$ and $\nabla^2\Phi = H \succ 0$. The VI minimizes the strictly convex Φ over K.

$$\widetilde{F}(x^*) = 0 \quad \Longrightarrow \quad \boxed{x_1^* = x_2^* = 2}.$$

This feasible zero proves existence. The equilibrium is unique.

Example 2: different nonlinear best responses

Two players choose $x_i \geq 0$, with $K = \mathbb{R}^2_+$.

Payoffs

$$u_1(x) = x_1(1 + x_2) - x_1^2 - \tfrac{1}{3}x_1^3,$$
$$u_2(x) = x_2 \ln(2 + x_1) - \tfrac{1}{2}x_2^2.$$

Each payoff is strictly concave in the player's own action.

Best responses

$$B_1(x_2) = \sqrt{2 + x_2} - 1, \qquad B_2(x_1) = \ln(2 + x_1).$$

Every equilibrium is interior because both best responses are positive on K.

Example 2: a nonlinear sign-equivalent transformation

The negative marginal payoffs define $F_i = -\partial_i u_i$.

Original VI

$$F(x) = \begin{pmatrix} x_1^2 + 2x_1 - 1 - x_2 \\ x_2 - \ln(2 + x_1) \end{pmatrix}.$$

$$\partial_2 F_1 = -1, \quad \partial_1 F_2 = -\frac{1}{2 + x_1}.$$

After SET

$$\widetilde{F}(x) = \begin{pmatrix} x_1^2 + 2x_1 - 1 - x_2 \\ e^{x_2} - 2 - x_1 \end{pmatrix}.$$

$$\partial_2 \widetilde{F}_1 = \partial_1 \widetilde{F}_2 = -1.$$

The exponential is strictly increasing:

$$x_2 \gtreqless \ln(2 + x_1) \quad \Longleftrightarrow \quad e^{x_2} \gtreqless 2 + x_1.$$

All component signs agree on the rectangular domain K. Thus

$$\mathrm{NE} = \mathrm{Sol}(K, F) = \mathrm{Sol}(K, \widetilde{F}).$$

Example 2: a convex equilibrium problem

The transformed operator is a gradient, $\widetilde{F} = \nabla\Phi$, where

$$\Phi(x) = \tfrac{1}{3}x_1^3 + x_1^2 - x_1 - x_1x_2 + e^{x_2} - 2x_2.$$

For every $x \in K$,

$$\nabla^2\Phi(x) = \begin{pmatrix} 2 + 2x_1 & -1 \\ -1 & e^{x_2} \end{pmatrix} \succeq \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \succ 0.$$

Unique equilibrium

Strong convexity and coercivity give a unique minimizer on K.

SET exposes a convex optimization representation of the same equilibrium problem.

Applications of general interest

1. **Pricing and social interactions:** oligopoly with price bounds and nonlinear peer effects.
2. **Nonlinear network complements:** when equilibrium exists, who acts most, and who is the key player?
3. **Network public goods:** how convex costs change participation, uniqueness, and effort rankings.

Only one methodological fact is needed before the examples.

The one methodological fact we need

Sign-equivalent transformation

If every component of F and $\widetilde{F}$ has the same sign at every feasible profile, then on a product of individual intervals the two variational inequalities have exactly the same solution set.

Player's choice	Equilibrium condition	What matters
Lower bound	$F_i \geq 0$	Sign
Interior	$F_i = 0$	Zero
Upper bound	$F_i \leq 0$	Sign

The product-domain condition matters when a choice is at a boundary.

What a better representation can reveal

P-property route

A transformed derivative matrix can establish at most one equilibrium. **Used for:** oligopoly pricing, peer effects, CES complements.

Potential route

A symmetric transformed system can become an optimization problem; curvature can give uniqueness. **Used for:** network games, public goods, multiplex interactions.

Existence needs its own argument in each application.

Oligopoly: what price should each firm charge?

There are $n \geq 2$ firms, each selling one differentiated product.

- Firm i has constant unit cost c_i and chooses $p_i \in [a_i, b_i]$, with $a_i \geq c_i$.
- Demand $Q_i(p) > 0$ depends on its own price and its rivals' prices.
- Consumers can also choose an outside option, whose price p_0 is fixed.

The firm's problem

$$\max_{p_i \in [a_i, b_i]} \pi_i(p) = (p_i - c_i) Q_i(p).$$

Can firms' pricing decisions be mutually consistent in more than one way, including when a price floor or cap binds?

Where demand comes from

Each consumer chooses the option with the highest utility:

$$U_i = v_i - p_i + \varepsilon_i, \qquad i = 0, 1, \ldots, n.$$

Here v_i measures product appeal; ε_i captures individual tastes. Option 0 means not buying from any firm. Normalize the mass of consumers to one.

A familiar example: multinomial logit

With independent standard extreme-value taste shocks,

$$Q_i(p) = \frac{e^{v_i - p_i}}{e^{v_0 - p_0} + \sum_{j=1}^{n} e^{v_j - p_j}}.$$

A higher own price loses customers to rivals or the outside option. The paper's argument covers more than this logit example.

An optimal price balances two effects

Differentiate profit, holding rivals' prices fixed:

$$\frac{\partial \pi_i}{\partial p_i} = \underbrace{Q_i}_{\text{more revenue per customer}} + \underbrace{(p_i - c_i)\frac{\partial Q_i}{\partial p_i}}_{\text{lost customers}}, \qquad \frac{\partial Q_i}{\partial p_i} < 0.$$

At an interior optimum

$$\underbrace{p_i - c_i}_{\text{actual markup}} = \underbrace{\frac{Q_i}{-\partial Q_i / \partial p_i}}_{\text{demand-implied markup } \mu_i(p)}.$$

μ_i is the inverse own-price semi-elasticity, $1/[-\partial_{p_i} \log Q_i]$. Less price-sensitive demand supports a larger markup; rivals' prices affect this sensitivity.

SET: rewrite pricing as a markup gap

The original VI uses negative marginal profit, $F_i = -\partial \pi_i / \partial p_i$. Divide by the positive number $-\partial Q_i / \partial p_i$:

A sign-equivalent pricing condition

$$\widetilde{F}_i(p) = \frac{F_i(p)}{-\partial Q_i / \partial p_i} = \underbrace{p_i - c_i}_{\text{actual markup}} - \underbrace{\mu_i(p)}_{\text{desired markup}}.$$

- $\widetilde{F}_i < 0$: the firm wants to raise its price.
- $\widetilde{F}_i > 0$: the firm wants to cut its price.
- $\widetilde{F}_i = 0$: the two pricing effects balance.

This is a new representation of the marginal condition, not a new profit function.

Price floors and caps: signs, not just zeros

Chosen price	Marginal profit	Markup gap
Floor: $p_i = a_i$	$\partial \pi_i / \partial p_i \leq 0$	$\widetilde{F}_i \geq 0$
Interior	$\partial \pi_i / \partial p_i = 0$	$\widetilde{F}_i = 0$
Cap: $p_i = b_i$	$\partial \pi_i / \partial p_i \geq 0$	$\widetilde{F}_i \leq 0$

Under E1 (own-price log-concavity), these conditions characterize best responses. Positive rescaling preserves all three cases. On $K = \prod_i [a_i, b_i]$,

$$\mathrm{NE} = \mathrm{Sol}(K, F) = \mathrm{Sol}(K, \widetilde{F}).$$

The entire equilibrium set is preserved. Looking only for zeros would miss equilibria with binding bounds.

The demand restrictions behind uniqueness

Write $\ell_i = \log Q_i$. Demand is positive, smooth, and decreasing in own price.

1. **E1: own-price log-concavity**
$$\frac{\partial^2 \ell_i}{\partial p_i^2} \leq 0.$$
2. **E2: cross-price log-supermodularity**
$$\frac{\partial^2 \ell_i}{\partial p_i \partial p_j} \geq 0 \quad (j \neq i, \text{ including the outside option } j = 0).$$
3. **E3: common-price-shift invariance**
$$Q_i(p_1 + t, \ldots, p_n + t, p_0 + t) = Q_i(p_1, \ldots, p_n, p_0).$$

The logit example satisfies all three. The outside price is fixed during the game; E3 describes demand when *all* option prices shift together.

Why the transformed pricing system is unique

Let $J = D\widetilde{F}$ be the derivative matrix with respect to firms' prices. The demand restrictions imply

$$J_{ii} \geq 1, \qquad J_{ij} \leq 0 \ (j \neq i), \qquad J_{ii} - \sum_{j \neq i} |J_{ij}| \geq 1.$$

An own-price change affects the markup gap more than all rival-price effects combined.

The P-property route

Positive diagonal and strict row diagonal dominance give the P-structure. The transformed VI, and hence the pricing game, has at most one equilibrium.

For finite price intervals, continuity and own-price quasi-concavity also give existence: the equilibrium is then unique.

A two-firm logit example: solve for prices

Take two identical firms: $c_1 = c_2 = v_1 = v_2 = 0$ and $v_0 - p_0 = 0$.

$$Q_i(p) = \frac{e^{-p_i}}{1 + e^{-p_1} + e^{-p_2}}, \qquad -\frac{\partial Q_i}{\partial p_i} = Q_i(1 - Q_i).$$

The markup gap becomes especially simple

$$\mu_i = \frac{1}{1 - Q_i}, \qquad \widetilde{F}_i = p_i - \frac{1}{1 - Q_i}.$$

With $p_i \in [0, 4]$, look for a symmetric interior price p:

$$p = \frac{1 + 2e^{-p}}{1 + e^{-p}} \quad \Longrightarrow \quad \boxed{p_1^* = p_2^* \approx 1.227, \quad Q_1^* = Q_2^* \approx 0.185.}$$

The uniqueness result also rules out asymmetric equilibria.

The same market with price floors or caps

Keep demand and costs unchanged; change only the feasible price intervals.

Interval for both firms	Equilibrium prices	Markup gap
$[0, 4]$: no binding bound	$(1.227, 1.227)$	0
$[0, 1]$: cap at 1	$(1, 1)$	-0.269
$[1.5, 4]$: floor at 1.5	$(1.5, 1.5)$	$+0.318$

At the cap firms want higher prices; at the floor they want lower prices.

What SET buys us

Bounds change prices, but sign preservation lets the same uniqueness argument cover all three cases.

Existing application: nonlinear peer effects

A social norm need not be the average peer

$$\widetilde{x}_{-i}(\beta) = \left(\sum_j \widetilde{g}_{ij} x_j^{\beta} \right)^{1/\beta}, \qquad x_i = (1 - \lambda_2) a_i + (\lambda_1 + \lambda_2) \widetilde{x}_{-i}(\beta).$$

- $\beta = 1$ recovers linear-in-means; other values change whose behavior is most influential.
- Taking logs exposes a *P*-property: under the paper's parameter restrictions, any solution is unique.
- The result is conditional on existence; the transformation alone does not supply it.

Existing applications: network games and multiplexity

Network public goods with income

An inverse Engel curve turns heterogeneous marginal conditions into a symmetric system on an undirected graph. A potential and a spectral condition can then give uniqueness.

Several network layers

An agent chooses effort across multiple kinds of ties. A transformed system can have a potential even with cross-layer costs. Extra curvature conditions are needed for uniqueness.

The same method can reveal different mathematical structures across models.

CES peer effects on an unnormalized network

$$u_i(x) = a_i x_i - \frac{1}{2}x_i^2 + \delta x_i \Big(\sum\nolimits_j g_{ij} x_j^{\beta}\Big)^{1/\beta}.$$

Best response

$$T_i(x) = a_i + \delta S_i(x), \qquad S_i(x) = \Big(\sum\nolimits_j g_{ij} x_j^{\beta}\Big)^{1/\beta}.$$

- $a_i > 0$, $\delta > 0$, $\beta > 0$, and $x_i \geq 0$.
- G is an undirected adjacency matrix, without self-loops.
- We use a connected, nonempty network for the Perron-limit results.

What the CES parameter changes

For positive neighbor actions,

$$\frac{\partial S_i}{\partial x_j} = g_{ij} x_j^{\beta-1} S_i(x)^{1-\beta}.$$

For two linked neighbors j and k,

$$\frac{\partial S_i / \partial x_j}{\partial S_i / \partial x_k} = \left(\frac{x_j}{x_k}\right)^{\beta-1}.$$

Relative peer influence

$\beta > 1$ favors high-action peers at the margin. $0 < \beta < 1$ places relatively more marginal weight on low-action peers.

Changing β also changes the scale of this unnormalized aggregate.

CES complements: uniqueness without a closed form

Every equilibrium has $x_i > 0$. Write $z_i = \log x_i$ and compare the positive quantities in the best-response equation:

$$H_i(z) = z_i - \log(a_i + \delta S_i(e^z)) .$$

What the transformation reveals

$$DH(z) = I - W(z), \qquad W(z) \geq 0, \qquad \sum_j W_{ij}(z) = \frac{\delta S_i}{a_i + \delta S_i} < 1.$$

The positive baseline a_i leaves some own-action weight in every row. Hence there is at most one finite positive equilibrium.

Whether a finite equilibrium exists depends on the network threshold on the next slide.

The spectral existence threshold

Theorem 4

For $\bar{\delta} = \lambda_{\max}(G)^{-1/\beta}$:

- $0 < \delta < \bar{\delta}$: a unique positive equilibrium exists.
- $\delta \uparrow \bar{\delta}$: all actions diverge, with convergent ratios.
- $\delta \geq \bar{\delta}$: no finite positive equilibrium exists.

On a d-regular network with $a_i = a$,

$$x_i^* = \frac{a}{1 - \delta d^{1/\beta}}, \qquad \bar{\delta} = d^{-1/\beta}.$$

Comparative statics below the threshold

How shocks change behavior

Within the region where a finite equilibrium exists,

$$a' \geq a \quad \Longrightarrow \quad x^*(a', \delta, \beta) \geq x^*(a, \delta, \beta),$$

$$\delta' \geq \delta \quad \Longrightarrow \quad x^*(a, \delta', \beta) \geq x^*(a, \delta, \beta).$$

- A higher individual baseline or stronger peer interaction raises equilibrium effort across the network.
- Equilibrium payoff is $u_i(x^*) = \frac{1}{2}(x_i^*)^2$: changes in effort also change welfare.

Bonacich centrality as the linear special case

When $\beta = 1$,

$$x = a + \delta G x.$$

The network multiplier

$$(I - \delta G)x = a, \qquad x^* = (I - \delta G)^{-1} a.$$

- $a = \mathbf{1}$: a standard Katz–Bonacich centrality vector.
- Heterogeneous a: baseline-weighted centrality.
- In the notation $(I - \delta G)x = b$, the vector b is this same baseline.

Eigenvector centrality at the critical limit

For the positive unit Perron vector e,

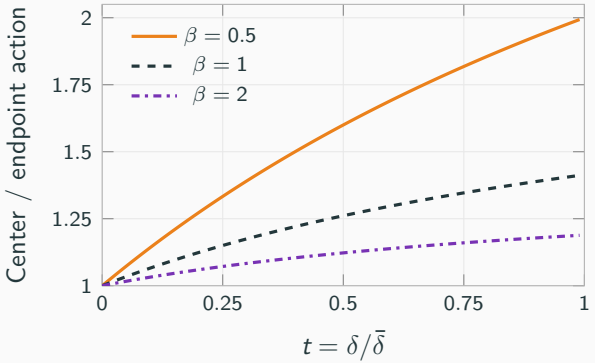
$$Ge = \lambda_{\max}(G)e, \qquad e \gg 0.$$

Theorem 4(ii)

$$\lim_{\delta \uparrow \bar{\delta}} \frac{x_i^*}{x_j^*} = \left(\frac{e_i}{e_j}\right)^{1/\beta}.$$

- $\beta = 1$: $x^*/\|x^*\|_2 \to e$.
- General $\beta > 0$: the vector $(x^*)^\beta$ approaches the Perron direction.
- Baseline heterogeneity affects levels, but disappears from these limiting ratios.

Relative actions along the path



$$\frac{C}{L} = \frac{1 + tq}{1 + t/q}, \qquad q = 2^{1/(2\beta)}.$$

As $t \uparrow 1$

The ratio converges to q.

The action levels diverge.

At the same fraction of the threshold, smaller β produces a larger center advantage here.

The key-player intervention

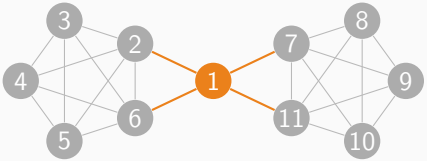
Let G^{-i} denote the network after removing player i.

Total equilibrium activity lost

$$\Delta_i = \sum_j x_j^*(G) - \sum_{j \neq i} x_j^*(G^{-i}), \qquad i^* \in \arg\max_i \Delta_i.$$

- Removing a player eliminates their action and changes everyone else's incentives.
- Highest equilibrium action need not identify the largest total effect.
- For general β, recompute equilibrium on each remaining network.

A bridge can matter more than the most active player



Linear benchmark, $\beta = 1$

δ	Key-player type
0.1	Node 2
0.2	Node 1

Nodes 2, 6, 7, and 11 have the highest degree and the highest action.

Nonlinear peers change the intervention ranking

At $\delta = 0.1$, the paper finds a switch from node 1 to the node-2 type near $\beta = 0.722$ within its reported parameter range.

Network public goods with nonlinear costs

On an undirected network, player i chooses $x_i \geq 0$ and receives

$$u_i(x) = b\left(x_i + \delta \sum_j g_{ij} x_j\right) - c(x_i), \qquad 0 < \delta \leq 1.$$

Assumption 2: benefits, costs, and a marginal crossing

$$b' > 0, \qquad b'' < 0, \qquad c'(x) > 0 \ (x > 0), \qquad c'' \geq 0,$$

$$b'(0) > c'(0), \qquad \lim_{x \to \infty} b'(x) < \lim_{x \to \infty} c'(x).$$

The positive autarky effort k^* uniquely solves $b'(k^*) = c'(k^*)$.

The benchmark of Bramoullé and Kranton (2007) has $c(x) = x$ and $\delta = 1$.

Best responses, corners, and existence

Let $t_i = \delta \sum_j g_{ij} x_j$ be the public good supplied by neighbors.

If $x_i > 0$

The agent balances marginal benefit and cost:

$$b'(x_i + t_i) = c'(x_i).$$

If $x_i = 0$

Neighbors may already provide enough that contributing the first unit is not worthwhile.

More provision by neighbors weakly reduces one's best response. Every best response is at most the autarky effort k^*, so a finite equilibrium exists.

Substitutes: curvature can restore uniqueness

The original marginal conditions are asymmetric because each agent receives different total provision. A sign-equivalent inverse transformation gives

$$\widetilde{F}_i(x) = x_i + \delta \sum_j g_{ij} x_j - h(x_i), \qquad h = (b')^{-1} \circ c'.$$

Economic result

On an undirected graph this system has a potential. The sufficient condition

$$1 + \delta \lambda_{\min}(G) > 0$$

ensures a unique equilibrium. Convex private costs add curvature and can support uniqueness beyond this simple bound.

The sign equivalence also retains equilibria with zero effort.

Zero marginal cost at zero implies full participation

Now suppose c is strictly convex, $c'(0) = 0$, and $\delta = 1$, as in Proposition 7.

Every equilibrium is interior

At any finite profile with $x_i = 0$,

$$\left.\frac{\partial u_i}{\partial x_i}\right|_{x_i=0} = b'\Big(\sum\nolimits_j g_{ij}x_j\Big) - c'(0) > 0.$$

Hence $x_i^* > 0$ for every player at every Nash equilibrium.

All equilibrium conditions therefore hold with equality:

$$b'\Big(x_i^* + \sum\nolimits_j g_{ij}x_j^*\Big) = c'(x_i^*).$$

Full participation does not require equilibrium uniqueness.

Complete networks: convex costs remove the continuum

Take the complete network and $\delta = 1$. Every player receives total provision $S = \sum_j x_j$.

Strictly convex costs, $c'(0) = 0$

$$c'(x_i) = b'(S) \quad \text{for all } i.$$

Strictly increasing c' forces equal effort.

$$x_i = \bar{x}, \qquad b'(n\bar{x}) = c'(\bar{x}).$$

The equilibrium is unique.

Linear costs, $c(x) = x$

With $b'(k^*) = 1$, every profile satisfying

$$x_i \geq 0, \qquad \sum_i x_i = k^*$$

is an equilibrium.

The continuum includes both equal sharing and a single specialist.

Here $\lambda_{\min}(G) = -1$. Uniqueness with strictly convex costs holds even though $1 + \delta\lambda_{\min}(G) = 0$.

Network structure determines effort rankings

Maintain homogeneous b, c, strictly convex costs, $c'(0) = 0$, and $\delta = 1$. Let N_i and $\overline{N}_i = N_i \cup \{i\}$ denote open and closed neighborhoods.

Neighborhood inclusion

At every equilibrium,

$$N_i \subsetneq N_j \text{ or } \overline{N}_i \subsetneq \overline{N}_j \implies x_i^* > x_j^*.$$

- **Star, $n \geq 3$.** Leaves choose equal effort. The hub contributes less.
- **Nested split graph.** If $d_i < d_j$, then $x_i^* > x_j^*$.

Degree alone does not imply this ranking on arbitrary networks.

What SET establishes in these applications

Question	Required argument
Equilibrium equivalence	Valid VI representation, full-domain sign equivalence, and appropriate domain geometry.
Existence	A compactness, coercivity, strong-P, or bounded-iteration argument.
Uniqueness	A suitable P property, strict monotonicity, or strict convexity.

Main conclusions

A general method

On rectangular domains, the signs of the equilibrium operator determine the VI solution set. Suitable transformations can reveal structure absent from the original Jacobian.

- CES complements combine a SET uniqueness argument with a Perron existence threshold.
- Bonacich centrality is a resolvent response. Eigenvector centrality describes its critical direction.
- Under strategic substitutes, cost curvature and the smallest network eigenvalue govern uniqueness.

Which transformation exposes the useful structure in a given model?

Backup: the oligopoly P-matrix calculation

Let $\ell_{i,j} = \partial \log Q_i / \partial p_j$ and $\ell_{i,jk} = \partial^2 \log Q_i / (\partial p_j \partial p_k)$. Then

$$\widetilde{F}_i = p_i - c_i + \frac{1}{\ell_{i,i}}, \qquad J_{ij} = \mathbf{1}_{\{i=j\}} - \frac{\ell_{i,ij}}{(\ell_{i,i})^2}.$$

E1 gives $J_{ii} \geq 1$; E2 gives $J_{ij} \leq 0$ for $j \neq i$. Differentiate E3's common-shift identity, then differentiate with respect to p_i:

$$\sum_{j=0}^{n} \ell_{i,j} = 0 \quad \Longrightarrow \quad \sum_{j=0}^{n} \ell_{i,ij} = 0.$$

Since E2 also gives $\ell_{i,i0} \geq 0$,

$$J_{ii} - \sum_{j \neq i} |J_{ij}| = \sum_{j=1}^{n} J_{ij} = 1 + \frac{\ell_{i,i0}}{(\ell_{i,i})^2} \geq 1.$$

Every principal submatrix has positive diagonal and strict row diagonal dominance, so every principal determinant is positive. Thus J is a P-matrix. The uniform dominance also gives the P-property of $\widetilde{F}$ on the price box.

Backup: logit checks and the role of SET

For unit-slope logit, let Q_0 denote the outside-option share. Then

$$\ell_{i,i} = -(1 - Q_i), \qquad \ell_{i,ii} = -Q_i(1 - Q_i) \le 0, \qquad \ell_{i,ij} = Q_i Q_j \ge 0 \quad (j \ne i,\ j = 0 \text{ allowed}).$$

Adding t to every option's price multiplies every logit weight by e^{-t}, so shares are unchanged. Thus E1–E3 hold.

The original derivative need not be a P-matrix

In the two-firm example, take $p_1 = p_2 = 4$ and write $s = Q_1 = Q_2$. For $F_i = -\partial \pi_i / \partial p_i$,

$$\frac{\partial F_i}{\partial p_i} = s(1 - s)\big[2 - p_i(1 - 2s)\big] < 0.$$

A negative 1×1 principal minor rules out a global P-matrix test for DF. The transformed derivative passes the test throughout the same price box.

E1 makes profit log-concave for $p_i > c_i$, not necessarily concave: $\log \pi_i = \log(p_i - c_i) + \log Q_i$.

Backup: a formal coordinate proof of Theorem 1

Let $K = \prod_i K_i$ and suppose $x^* \in \mathrm{Sol}(K, F)$. For every i and $t \in K_i$,

$$(t - x_i^*) F_i(x^*) \geq 0.$$

Sign equivalence implies the same inequality for $\widetilde{F}_i(x^*)$:

$$\begin{cases} F_i(x^*) = 0 & \Longrightarrow \ \widetilde{F}_i(x^*) = 0, \\ F_i(x^*) > 0 & \Longrightarrow \ t - x_i^* \geq 0 \text{ and } \widetilde{F}_i(x^*) > 0, \\ F_i(x^*) < 0 & \Longrightarrow \ t - x_i^* \leq 0 \text{ and } \widetilde{F}_i(x^*) < 0. \end{cases}$$

Consequently, for every $x \in K$,

$$\langle x - x^*, \widetilde{F}(x^*) \rangle = \sum_i (x_i - x_i^*) \widetilde{F}_i(x^*) \geq 0.$$

Thus $\mathrm{Sol}(K, F) \subseteq \mathrm{Sol}(K, \widetilde{F})$. Interchanging the two operators gives the reverse inclusion.

Backup: matrix tests in the payoff-scaling example

For $J = \begin{pmatrix} 1 & -\delta \\ -\kappa\delta & \kappa \end{pmatrix}$ with $\kappa > 0$:

P-matrix test

The principal minors are

$$1, \quad \kappa, \quad \kappa(1-\delta^2).$$

All are positive if $0 < \delta < 1$.

Strict monotonicity test

For $S = (J + J^\top)/2$,

$$\det S = \kappa - \frac{(1+\kappa)^2\delta^2}{4}.$$

$S \succ 0$ iff $\delta < 2\sqrt{\kappa}/(1+\kappa)$.

Let $\Delta = (1-\kappa)^2 + (1+\kappa)^2\delta^2$. The smallest eigenvalue is

$$\lambda_{\min}(S) = \frac{1+\kappa-\sqrt{\Delta}}{2}.$$

Rescaling by $\mathrm{diag}(1, 1/\kappa)$ restores the symmetric derivative with eigenvalues $1-\delta$ and $1+\delta$.

Backup: the nonlinear Perron limit

Let $y(\delta) = x(\delta)^\beta$ and normalize $q(\delta) = y(\delta)/\|y(\delta)\|_1$.

- If actions remained bounded as $\delta \uparrow \bar{\delta}$, continuity would give an equilibrium at $\bar{\delta}$.
- Connectedness propagates divergence: $x_i \geq \delta x_j$ whenever $g_{ij} = 1$.

Therefore $a_i/x_i \to 0$, and along any convergent subsequence,

$$\frac{(x-a)^\beta}{\|x^\beta\|_1} = \delta^\beta G q(\delta) \quad \Longrightarrow \quad q = \bar{\delta}^\beta G q.$$

Uniqueness of the normalized limit

The only nonnegative unit-ℓ_1 Perron vector is $e/\|e\|_1$. Every subsequential limit is the same, so the full sequence converges.

Backup: algebraic solutions and positive equilibria

For two linked agents with baseline $a = \mathbf{1}$,

$$G = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \lambda_1 = 1, \quad \lambda_2 = -1.$$

Whenever $\delta \neq \pm 1$, the linear system has the solution

$$x = (I - \delta G)^{-1}\mathbf{1} = \frac{1}{1-\delta}\mathbf{1}.$$

Parameter	Algebraic system	Economic equilibrium
$0 < \delta < 1$	Unique solution	Positive and finite
$\delta = 1$	Singular, inconsistent	None
$\delta > 1$	Unique negative solution	None on $\mathbb{R}^2_+$

Backup: complete-network uniqueness without the strict spectral bound

Assume homogeneous b, c, strictly convex c, $c'(0) = 0$, and $\delta = 1$. On the complete network, write $S = \sum_j x_j$.

1. Full participation makes every first-order condition an equality:
$$c'(x_i) = b'(S) = c'(x_j) \qquad \text{for every } i, j.$$
2. Strictly increasing c' implies $x_i = x_j = \bar{x}$.
3. The function $f(z) = b'(nz) - c'(z)$ is strictly decreasing, with $f(0) > 0$ and an eventually negative value. Thus it has exactly one positive zero.

Why the spectral test is only sufficient

For $n \geq 2$, the complete graph has $\lambda_{\min}(G) = -1$. At $\delta = 1$ the strict spectral inequality fails, but the argument above proves uniqueness.

Backup: why the hub contributes less in a star

Let $n \geq 3$ and $\delta = 1$, with homogeneous functions, strictly convex costs, and $c'(0) = 0$. Write x_H for the hub's effort.

1. Every leaf faces the same peer effort x_H and has a unique best response. Thus all leaves choose one common effort $x_L > 0$.
2. The hub and a leaf satisfy
$$c'(x_H) = b'(x_H + (n-1)x_L), \qquad c'(x_L) = b'(x_H + x_L).$$
3. Since $(n-2)x_L > 0$ and b' is strictly decreasing,
$$c'(x_H) < c'(x_L).$$
Strictly increasing c' then gives $\boxed{x_H < x_L}$.

The proof applies to every equilibrium, without a uniqueness assumption.

References

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The seminar follows the supplied SET draft. The two-player walkthroughs, two-firm logit calculations, path-network curves, and two-node calculation are seminar examples.